

CHE 312 HW 11

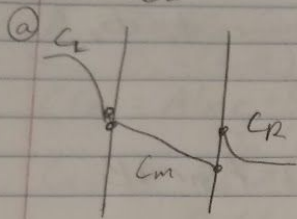
1.

$C_L > C_R$

$\phi = \frac{C_m}{C_L}$

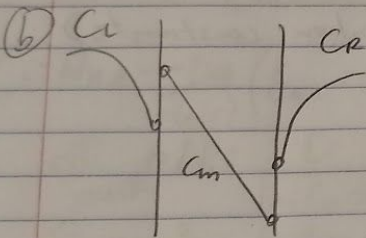
$\phi = \frac{C_m}{C_R}$

$>, <, \text{ or } = 1?$



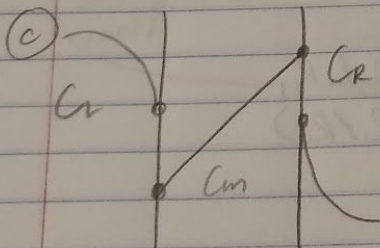
Yes, it is physically possible, as the concentration drops in the middle and starts lower at C_R and drops rapidly as it goes on.

$\phi = \frac{C_m}{C_L} < 1$ $\phi = \frac{C_m}{C_R} < 1$



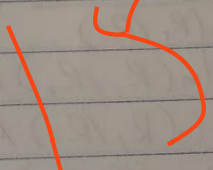
Yes, the concentration profile is possible. The middle affects the C_L & C_R by dropping the concentration, then it increases rapidly.

$\phi = \frac{C_m}{C_L} > 1$ $\phi = \frac{C_m}{C_R} < 1$

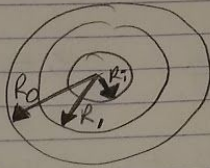


Yes, the profile is possible, as the membrane affects the two concentrations as they get closer & further to it.

$\phi = \frac{C_m}{C_L} < 1$ $\phi = \frac{C_m}{C_R} > 1$



2. Artery w/:
- | | |
|-------------|-------|
| $R_0 - R_i$ | D_0 |
| $R_i - R_j$ | D_i |



Find effective diffusivity term

Assume $\phi_1 = \phi_2$

Fick's law:

$$D \frac{1}{r} \left(\frac{d}{dr} \left(r \frac{dC}{dr} \right) \right) = -\phi$$

$$r \frac{dC}{dr} = A \Rightarrow \frac{dC}{dr} = \frac{A}{r} \Rightarrow C(r) = A \ln r + B$$

BC1: $C(R_i) = C_i = C_1$ w/ (A, B_1 as integration constants)

BC2: $C(R_0) = C_0 = C_2$ w/ (A_2, B_2 as integration constants)

BC3: Set $C_1 = C_2$

BC4: 1st derivative: $-D_1 \frac{dC_1}{dr} = -D_2 \frac{dC_2}{dr}$

diffusion

$$C(r) = A \ln r + B$$

$$C_i = A_1 \ln R_i + B_1 \rightarrow B_1 = C_i - A_1 \ln R_i$$

$$C_0 = A_2 \ln R_0 + B_2 \rightarrow B_2 = C_0 - A_2 \ln R_0$$

$$B_1 = C_i \quad C_i - A_1 \ln R_i = C_0 - A_2 \ln R_0$$

$B_2 = C_0$ $C_i - A_1 \ln R_i + A_1 \ln R_i = C_0 - A_2 \ln R_0 + A_2 \ln R_i$

$$C_i + A_1 (\ln R_i / R_i) = C_0 + A_2 \ln (R_i / R_0)$$

Solve for A_1 :

$$A_1 = \frac{C_0 - C_i + A_2 \ln (R_i / R_0)}{\ln (R_i / R_i)}$$

$$\frac{A_1 D_1}{R_i} = \frac{A_2 D_2}{R_1} \Rightarrow \frac{C_0 - C_i + A_2 \ln (R_i / R_0)}{\ln (R_i / R_i)} \frac{D_1}{R_i} = A_2 \frac{D_2}{R_1}$$

Solve for A_2

$$(C_0 - C_i + A_2 \ln (R_i / R_0)) D_1 = D_2 \ln (R_i / R_i) A_2$$

$$D_1 C_0 - D_1 C_i + D_1 A_2 \ln (R_i / R_0) = D_2 \ln (R_i / R_i) A_2$$

$$D_1 C_0 - D_1 C_i = -D_1 A_2 \ln (R_i / R_0) + D_2 \ln (R_i / R_i) A_2$$

$$D_1 C_0 - D_1 C_i = A_2 (D_2 \ln (R_i / R_i) - D_1 \ln (R_i / R_0))$$

$$A_2 = \frac{D_1 (C_0 - C_i)}{D_2 \ln (R_i / R_i) - D_1 \ln (R_i / R_0)}$$

then solve for A_1

$$A_1 = \frac{C_0 - C_i + A_2 \ln(R_1/R_0)}{\ln(R_1/R_i)} = \frac{C_0 - C_i + \left(\frac{D_1(C_0 - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_0)} \right) \ln(R_1/R_0)}{\ln(R_1/R_i)} \rightarrow$$

$$A_1 = \frac{D_2(C_0 - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_0)}$$

$$C(r) = A \ln r + B$$

$$C_1(r) = \frac{D_2(C_0 - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_0)} \cdot \ln\left(\frac{r}{R_i}\right) + C_i$$

$$C_2(r) = \frac{D_1(C_0 - C_i)}{D_2 \ln(R_1/R_i) - D_1 \ln(R_1/R_0)} \cdot \ln\left(\frac{r}{R_0}\right) + C_0$$

Table 14.1: $N_{\text{max}} = \frac{-DAB}{r \ln(r)} (C_{A,2} - C_{A,1}) \rightarrow N = \frac{-D}{\ln(R_0/R_i)} \cdot \frac{1}{r} (C_0 - C_i)$

eff. diff. coeff $= \frac{D}{\ln(R_0/R_i)}$

outer & inner terms

Since 3 cylinders:

$$\text{eff diff coeff} = \frac{D_1}{\ln(R_1/R_i)} + \frac{D_2}{\ln(R_0/R_1)}$$

AS